# Safety Data Sheet



## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

Antifreeze Extra

UFI: VUH3-Q9GF-A20A-SH78

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Use of the substance/mixture**

Refrigerant. Anti-freezing agent.

### 1.3. Details of the supplier of the safety data sheet

**Manufacturer**

| | | |
|---|---|---|
| Company name: | Emirates lubricants | |
| Street: | Ras Al khaimah | |
| Place: | United Arab Emirates | |
| Telephone: | +971 55 105 2851 | +971 52 660 6888 |
| e-mail: | info@emirateslubricant. | |
| Contact person: | Application Technology | |
| Internet: | www.emirateslubricant.com | |
| Responsible Department: | Emirates lubricants Application Technology | |

### 1.4. Emergency telephone number:

+971 52 660 6888 - This number is serviced during office hours.

## SECTION 2: Hazards identification

## 2.1. Classification of the substance or mixture

**GB CLP Regulation**

Acute Tox. 4; H302
Repr. 1B; H360D
STOT RE 2; H373

Full text of hazard statements: see SECTION 16.

The mixture is classified as hazardous according to regulation (EC) No 1272/2008 [CLP].

## 2.2. Label elements

**GB CLP Regulation**

**Hazard components for labelling**

ethanediol; ethylene glycol
Sodium 2-ethylhexanoate

**Signal word:** Danger

**Pictograms:**

**Hazard statements**

| | |
|---|---|
| H302 | Harmful if swallowed. |
| H360D | May damage the unborn child. |
| H373 | May cause damage to organs through prolonged or repeated exposure. |

**Precautionary statements**

| | |
|---|---|
| P102 | Keep out of reach of children. |
| P201 | Obtain special instructions before use. |
| P260 | Do not breathe dust/fume/gas/mist/vapours/spray. |
| P264 | Wash hands thoroughly after handling. |
| P280 | Wear protective gloves/protective clothing/eye protection/face protection/hearing protection. |
| P301+P310 | IF SWALLOWED: Immediately call a POISON CENTER/doctor. |
| P308+P313 | IF exposed or concerned: Get medical advice/attention. |
| P501 | Dispose of contents/container to an appropriate recycling or disposal facility. |

**Special labelling of certain mixtures**

Restricted to professional users.

**Additional advice on labelling**

Product is classified and labelled in accordance with EC regulations or the corresponding national laws.

## 2.3. Other hazards

Spilled product must not leak into the ground.

| CAS No | EC No | Chemical name | Quantity |
|---|---|---|---|
| | Specific Conc. Limits, M-factors and ATE | | |
| 107-21-1 | 203-473-3 | ethanediol; ethylene glycol | 80 - < 100 % |
| | inhalation: LC50 = >2,5 mg/l (vapours); dermal: LD50 = > 3500 mg/kg; oral: LD50 = 7712 mg/kg | | |
| 19766-89-3 | 243-283-8 | Sodium 2-ethylhexanoate | < 5 % |

| | | | |
|---|---|---|---|
| | dermal: LD50 = > 2000 mg/kg; oral: LD50 = 2043 mg/kg | | |
| 29385-43-1 | 249-596-6 | Methyl-1H-benzotriazole | 0.5 - < 1 % |
| | dermal: LD50 = > 2000 mg/kg; oral: LD50 = ca. 720 mg/kg | | |



Do not allow uncontrolled leakage of product into the environment.

## SECTION 3: Composition/information on ingredients

### 3.2. Mixtures

**Hazardous components**

| CAS No | Chemical name | | | Quantity |
|---|---|---|---|---|
| | EC No | Index No | REACH No | |
| | Classification (GB CLP Regulation) | | | |
| 107-21-1 | ethanediol; ethylene glycol | | | 80 - < 100 % |
| | 203-473-3 | 603-027-00-1 | | |
| | Acute Tox. 4, STOT RE 2; H302 H373 | | | |
| 19766-89-3 | Sodium 2-ethylhexanoate | | | < 5 % |
| | 243-283-8 | | 01-2119972937-17 | |
| | Repr. 1B; H360D | | | |
| 29385-43-1 | Methyl-1H-benzotriazole | | | 0.5 - < 1 % |
| | 249-596-6 | | 01-2119979081-35 | |
| | Repr. 2, Acute Tox. 4, Aquatic Chronic 2; H361d H302 H411 | | | |

Full text of H and EUH statements: see section 16.

**Specific Conc. Limits, M-factors and ATE**



**Further Information**

Classification system: The classification corresponds to the current EC lists and is completed by information from specialist literature and company information.

## SECTION 4: First aid measures

### 4.1. Description of first aid measures

**General information**

Self-protection of the first aider. Change contaminated clothing. Do not put any product-impregnated cleaning rags into your trouser pockets.

**After inhalation**

Move victim to fresh air. Put victim at rest and keep warm. In case of breathing difficulties administer oxygen. If victim is at risk of losing consciousness, position and transport on their side.

**After contact with skin**

After contact with skin, wash immediately with plenty of water and soap. In case of skin irritation, seek medical treatment.

**After contact with eyes**

In case of contact with eyes, rinse immediately with plenty of flowing water for 10 to 15 minutes holding eyelids apart. Subsequently consult an ophthalmologist.

**After ingestion**

Do NOT induce vomiting. Rinse mouth thoroughly with water. Call a physician immediately. Get medical advice/attention if you feel unwell.

### 4.2. Most important symptoms and effects, both acute and delayed

Ingestion causes nausea, weakness and central nervous system effects. The following symptoms may occur: Dizziness. Headache. drowsiness. Disorientation.

### 4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

## SECTION 5: Firefighting measures

### 5.1. Extinguishing media

**Suitable extinguishing media**

Extinguishing powder. Carbon dioxide ($CO_2$). alcohol resistant foam.

**Unsuitable extinguishing media**

High power water jet.

### 5.2. Special hazards arising from the substance or mixture

In case of fire may be liberated: Carbon monoxide Carbon dioxide ($CO_2$). ketone. aldehydes.

### 5.3. Advice for firefighters

In case of fire: Wear self-contained breathing apparatus.

**Additional information**

Co-ordinate fire-fighting measures to the fire surroundings. Use water spray jet to protect personnel and to cool endangered containers. In case of fire and/or explosion do not breathe fumes. Contaminated fire-fighting water must be collected separately. Do not allow to enter into surface water or drains.

## SECTION 6: Accidental release measures

### 6.1. Personal precautions, protective equipment and emergency procedures

**General advice**

High slip hazard because of leaking or spilled product. Remove all sources of ignition. Wear breathing apparatus if exposed to vapours/dusts/aerosols. Avoid contact with skin, eye and clothing.

### 6.2. Environmental precautions

Do not allow to enter into surface water or drains. In case of gas escape or of entry into waterways, soil or drains, inform the responsible authorities. Prevent spread over a wide area (e.g. by containment or oil barriers).

### 6.3. Methods and material for containment and cleaning up

**Other information**

Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents). Collect in closed containers for disposal. Clean contaminated articles and floor according to the environmental legislation.

### 6.4. Reference to other sections

Safe handling: see section 7
Personal protection equipment: see section 8
Section 12: Ecological Information (non-mandatory)
Disposal: see section 13

## SECTION 7: Handling and storage

### 7.1. Precautions for safe handling

**Advice on safe handling**

Keep away from sources of ignition - No smoking. The following must be prevented: inhalation. Provide adequate ventilation as well as local exhaustion at critical locations. Wear personal protection equipment. Avoid contact with skin, eyes and clothes.

**Advice on protection against fire and explosion**

In case of insufficient ventilation and/or through use, explosive/highly flammable mixtures may develop.

**Advice on general occupational hygiene**

Wash hands before breaks and after work. Take off immediately all contaminated clothing. Wash contaminated clothing prior to re-use. Do not eat, drink, smoke or sneeze at the workplace.

### 7.2. Conditions for safe storage, including any incompatibilities

**Requirements for storage rooms and vessels**

Keep the packing dry and well sealed to prevent contamination and absorbtion of humidity. Keep container tightly closed in a cool place. Keep/Store only in original container.

**Hints on joint storage**

Keep away from food, drink and animal feedingstuffs.
Keep away from: Oxidising agent, strong

**Further information on storage conditions**

Recommended storage temperature: 5 - 40°C
Protect against: heat. UV-radiation/sunlight. frost.

### 7.3. Specific end use(s)

Refrigerant. Anti-freezing agent. Further information: see technical data sheet.

## SECTION 8: Exposure controls/personal protection

### 8.1. Control parameters

**Exposure limits (EH40)**

| CAS No | Substance | ppm | mg/m³ | fibres/ml | Category | Origin |
|---|---|---|---|---|---|---|
| 107-21-1 | Ethane-1,2-diol, vapour | 20 | 52 | | TWA (8 h) | WEL |
| | | 40 | 104 | | STEL (15 min) | WEL |

**DNEL/DMEL values**

| CAS No | Substance | | | |
|---|---|---|---|---|
| DNEL type | | Exposure route | Effect | Value |
| 107-21-1 | ethanediol; ethylene glycol | | | |
| Worker DNEL, long-term | | inhalation | local | 35 mg/m³ |
| Worker DNEL, long-term | | dermal | systemic | 106 mg/kg bw/day |
| Consumer DNEL, long-term | | inhalation | local | 7 mg/m³ |
| Consumer DNEL, long-term | | dermal | systemic | 53 mg/kg bw/day |
| 19766-89-3 | Sodium 2-ethylhexanoate | | | |
| Worker DNEL, long-term | | inhalation | systemic | 14 mg/m³ |
| Worker DNEL, long-term | | dermal | systemic | 2 mg/kg bw/day |
| Consumer DNEL, long-term | | inhalation | systemic | 3,5 mg/m³ |
| Consumer DNEL, long-term | | dermal | systemic | 1 mg/kg bw/day |
| Consumer DNEL, long-term | | oral | systemic | 1 mg/kg bw/day |
| 29385-43-1 | Methyl-1H-benzotriazole | | | |
| Worker DNEL, long-term | | inhalation | systemic | 21,2 mg/m³ |
| Worker DNEL, long-term | | dermal | systemic | 0,3 mg/kg bw/day |
| Consumer DNEL, long-term | | dermal | systemic | 0,01 mg/kg bw/day |
| Consumer DNEL, long-term | | oral | systemic | 0,01 mg/kg bw/day |

**PNEC values**

| CAS No | Substance | |
|---|---|---|
| Environmental compartment | | Value |
| 19766-89-3 | Sodium 2-ethylhexanoate | |
| Freshwater | | 0,36 mg/l |
| Freshwater (intermittent releases) | | 0,493 mg/l |
| Marine water | | 0,036 mg/l |
| Freshwater sediment | | 0,301 mg/kg |
| Marine sediment | | 0,03 mg/kg |

| | | |
|---|---|---|
| Micro-organisms in sewage treatment plants (STP) | | 71,7 mg/l |
| Soil | | 0,058 mg/kg |
| 29385-43-1 | Methyl-1H-benzotriazole | |
| Freshwater | | 0,008 mg/l |
| Freshwater (intermittent releases) | | 0,086 mg/l |
| Marine water | | 0,02 mg/l |
| Freshwater sediment | | 0,117 mg/kg |
| Marine sediment | | 0,292 mg/kg |
| Micro-organisms in sewage treatment plants (STP) | | 39,4 mg/l |
| Soil | | 0,0187 mg/kg |

**Additional advice on limit values**

source:

## 8.2. Exposure controls




**Appropriate engineering controls**

Provide adequate ventilation as well as local exhaustion at critical locations.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection**

Tightly sealed safety glasses. German Industry Norms (DIN) / European Norms (EN): EN 166

**Hand protection**

Tested protective gloves are to be worn: German Industry Norms (DIN) / European Norms (EN): EN ISO 374

Duration of wearing with permanent contact: 480
min Suitable material: NBR (Nitrile rubber).
Thickness of glove material: 0.7 mm.

Wearing time with occasional contact (splashes): 30 min
Suitable material: NBR (Nitrile rubber).
Thickness of glove material: 0.4 mm

Protect skin by using skin protective cream.

**Skin protection**

Wear suitable protective clothing. Change contaminated clothing. Do not put any product-impregnated cleaning rags into your trouser pockets.

**Respiratory protection**

If technical exhaust or ventilation measures are not possible or insufficient, respiratory protection must be worn. Breathing protection with filter against organic gases and vapours type A - boiling point > 65°C: A1: < 1000 ppm; A2: < 5000 ppm; A3: < 10000 ppm

# Safety Data sheet

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| Physical state: | liquid | |
| Colour: | red - violet | |
| Odour: | characteristic | |
| Odour threshold: | not determined | |

| | | Test method |
|---|---|---|
| Melting point/freezing point: | not determined | |
| Boiling point or initial boiling point and boiling range: | 180 °C | |
| Flammability: | not determined | |
| Lower explosion limits: | not determined | |
| Upper explosion limits: | not determined | |
| Flash point: | 122 °C | DIN EN ISO 22719 |
| Auto-ignition temperature: | not determined | |
| Decomposition temperature: | not determined | |
| pH-Value: | 8,5 | |
| Viscosity / kinematic: | not applicable | |
| Water solubility: | easily soluble. | |

Solubility in other solvents
not determined

| | | |
|---|---|---|
| Partition coefficient n-octanol/water: | not determined | |



| | | |
|---|---|---|
| Vapour pressure: | not determined | |
| Density (at 20 °C): | 1,113 g/cm³ | DIN 51757 |
| Relative density: | not determined | |
| Relative vapour density: | not determined | |

### 9.2. Other information

**Information with regard to physical hazard classes**

Explosive properties
not determined

Self-ignition temperature

| | |
|---|---|
| Solid: | not determined |
| Gas: | not determined |

Oxidizing properties
not determined

## SECTION 10: Stability and reactivity

**10.1. Reactivity**

The product is stable under storage at normal ambient temperatures.

**10.2. Chemical stability**

The mixture is chemically stable under recommended conditions of storage, use and temperature. **10.3. Possibility of hazardous reactions** No known hazardous reactions.

**10.4. Conditions to avoid**

Do not overheat to avoid decomposition by heat.

Refer to chapter 7 No further action is necessary.

**10.5. Incompatible materials**

Reacts with: Oxidising agent, strong. Strong acid. Nitrates. Peroxides. Chlorates.

**10.6. Hazardous decomposition products**

In case of fire may be liberated: Carbon monoxide Carbon dioxide ($CO_2$). ketone. aldehydes.

## SECTION 11: Toxicological information

**11.1. Information on hazard classes as defined in GB CLP Regulation**

**Acute toxicity** Harmful
if swallowed.
Acute oral toxicity LD50: 1720 mg/kg (estimated)

**ATEmix calculated**

ATE (oral) 526,3 mg/kg

| CAS No | Chemical name | | | | |
|---|---|---|---|---|---|
| | Exposure route | Dose | Species | Source | Method |
| 107-21-1 | ethanediol; ethylene glycol | | | | |
| | oral | LD50 7712 mg/kg | Rat | Study report (1968) | according to BASF-internal standards |
| | dermal | LD50 > 3500 mg/kg | Mouse | Fundamental and Applied Toxicology 27: 1 | LD50 derived from developmental toxicity |
| | inhalation (4 h) vapour | LC50 >2,5 mg/l | Rat | | |
| 19766-89-3 | Sodium 2-ethylhexanoate | | | | |
| | oral | LD50 2043 mg/kg | Rat | Study report (1987) | OECD Guideline 401 |
| | dermal | LD50 > 2000 mg/kg | Rat | Study report (1986) | OECD Guideline 402 |
| 29385-43-1 | Methyl-1H-benzotriazole | | | | |
| | oral | LD50 ca. 720 mg/kg | Rat | Study report (1983) | OECD Guideline 401 |
| | dermal | LD50 > 2000 mg/kg | Rabbit | Study report (1984) | OECD Guideline 402 |

**Irritation and corrosivity**

Based on available data, the classification criteria are not met.

**Sensitising effects**

Based on available data, the classification criteria are not met.

**Carcinogenic/mutagenic/toxic effects for reproduction**

May damage the unborn child. (Sodium 2-ethylhexanoate)
Germ cell mutagenicity: Based on available data, the classification criteria are not met.
Carcinogenicity: Based on available data, the classification criteria are not met.

**STOT-single exposure**

Based on available data, the classification criteria are not met.

**STOT-repeated exposure**

May cause damage to organs through prolonged or repeated exposure. (ethanediol; ethylene glycol)
Prolonged/repetitive skin contact may cause skin defattening or dermatitis.

**Aspiration hazard**

Based on available data, the classification criteria are not met.

## 11.2. Information on other hazards

**Endocrine disrupting properties**

not applicable **Other information**

This product contains ethylene glycol (EG). The toxicity of EG via inhalation or skin contact is expected to be slight at room temperature. The estimated oral lethal dose is about 100 cc (3.3 oz.) for an adult human. Ethylene glycol is oxidized to oxalic acid which results in the deposition of calcium oxalate crystals mainly in the brain and kidneys. Early signs and symptoms of EG poisoning may resemble those of alcohol intoxication. Later, the victim may experience nausea, vomiting, weakness, abdominal and muscle pain, difficulty in breathing and decreased urine output. When EG was heated above the boiling point of water, vapors formed which reportedly caused unconsciousness, increased lymphocyte count, and a rapid, jerky movement of the eyes in persons chronically exposed. When EG was administered orally to pregnant rats and mice, there was an increase in fetal deaths and birth defects. Some of these effects occurred at doses that had no toxic effects on the mothers. We are not aware of any reports that EG causes reproductive toxicity in human beings.

## SECTION 12: Ecological information

### 12.1. Toxicity

Based on available data, the classification criteria are not met.
Mixture not tested.

| CAS No | Chemical name | | | | | |
|---|---|---|---|---|---|---|
| | Aquatic toxicity | Dose | [h] \| [d] | Species | Source | Method |
| 107-21-1 | ethanediol; ethylene glycol | | | | | |
| | Acute fish toxicity | LC50 53000 mg/l | 96 h | Pimephales promelas | Publication (1983) | other: ASTM Subcommittee on Safety to Aq |
| | Acute algae toxicity | ErC50 6500 - 13000 mg/l | 96 h | Raphidocelis subcapitata | Study report (1982) | other: EPA 600/9-78-018, 1978 |
| | Acute crustacea toxicity | EC50 > 100 mg/l | 48 h | Daphnia magna | Study report (1998) | OECD Guideline 202 |
| | Fish toxicity | NOEC > 40 mg/l | 28 d | Menidia peninsulae | Publication (1985) | other: ASTM E-47.01 |
| | Crustacea toxicity | NOEC 8590 mg/l | 7 d | Ceriodaphnia dubia | Environ. Toxicology and Chemistry, Vol. | other: EPA 600/4-89/001. U.S. Environmen |
| | Acute bacteria toxicity | (EC50 > 1000 mg/l) | 3 h | activated sludge, domestic | Chemosphere 14(9), 1239-1251 (1985) | OECD Guideline 209 |
| 19766-89-3 | Sodium 2-ethylhexanoate | | | | | |
| | Acute fish toxicity | LC50 > 100 mg/l | 96 h | Oryzias latipes | NITE (National Institute of Technology a | OECD Guideline 203 |
| | Acute algae toxicity | ErC50 49,3 mg/l | 72 h | Desmodesmus subspicatus | Study report (1988) | other: Method: other: German Industrial |
| | Acute crustacea toxicity | EC50 85,4 mg/l | 48 h | Daphnia magna | Study report (1988) | other: Directive 79/831/EEC, Annex V, Pa |
| | Crustacea toxicity | NOEC 25 mg/l | 21 d | Daphnia magna | Study report (1997) | OECD Guideline 211 |
| 29385-43-1 | Methyl-1H-benzotriazole | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Acute fish toxicity | LC50 | 55 mg/l | 96 h | Cyprinodon variegatus | Study report (2003) | other: The test procedure is based on te |
| | Acute algae toxicity | ErC50 | 75 mg/l | 72 h | Raphidocelis subcapitata | Study report (1994) | OECD Guideline 201 |
| | Acute crustacea toxicity | EC50 mg/l | 15,8 | 48 h | other aquatic crustacea: Daphnia galeata | Environ Sci Pollut Res 19:1781-1790 (201 | OECD Guideline 202 |
| | Crustacea toxicity | NOEC mg/l | 18,4 | 21 d | Daphnia magna | Study report (1995) | other: "Daphnia Reproduction Test" of OE |

### 12.2. Persistence and degradability

Product is partially biodegradable.

### 12.3. Bioaccumulative potential

No data available

**Partition coefficient n-octanol/water**

| CAS No | Chemical name | Log Pow |
|---|---|---|
| 107-21-1 | ethanediol; ethylene glycol | -1,36 |
| 19766-89-3 | Sodium 2-ethylhexanoate | 1,3 |
| 29385-43-1 | Methyl-1H-benzotriazole | 1,079 |

### 12.4. Mobility in soil

No data available

### 12.5. Results of PBT and vPvB assessment

The substances in the mixture do not meet the PBT/vPvB criteria.

### 12.6. Endocrine disrupting properties

This product does not contain a substance that has endocrine disrupting properties with respect to non-target organisms as no components meets the criteria.

### 12.7. Other adverse effects

No data available

**Further information**

Do not allow to enter into surface water or drains.

## SECTION 13: Disposal considerations

### 13.1. Waste treatment methods

**Disposal recommendations**

Must not be disposed of with domestic refuse. Do not allow to enter into surface water or drains.

**List of Wastes Code - residues/unused products**

160114 WASTES NOT OTHERWISE SPECIFIED IN THE LIST; end-of-life vehicles from different means of transport (including off-road machinery) and wastes from dismantling of end-of-life vehicles and vehicle maintenance (except 13, 14, 16 06 and 16 08); antifreeze fluids containing hazardous substances; hazardous waste

**Contaminated packaging**

Contaminated packages must be completely emptied and can be re-used following proper cleaning. Dispose of waste according to applicable legislation. Packing which cannot be properly cleaned must be thrown away.

## SECTION 14: Transport information

**Land transport (ADR/RID)**

**14.1. UN number or ID number:** -

**14.2. UN proper shipping name:** -

**14.3. Transport hazard class(es):** -

**14.4. Packing group:** -

**Inland waterways transport (ADN)**

**14.1. UN number or ID number:** -

**14.2. UN proper shipping name:** -

**14.3. Transport hazard class(es):** -

**14.4. Packing group:** -

**Marine transport (IMDG)**

**14.1. UN number or ID number:** -

**14.2. UN proper shipping name:** -

**14.3. Transport hazard class(es):** -

**14.4. Packing group:** -

**Air transport (ICAO-TI/IATA-DGR)**

**14.1. UN number or ID number:** -

**14.2. UN proper shipping name:** -

**14.3. Transport hazard class(es):** -

**14.4. Packing group:** -

**14.5. Environmental hazards**

ENVIRONMENTALLY HAZARDOUS: No

**14.6. Special precautions for user**

Unless specified otherwise, general measures for safe transport must be followed.

**14.7. Maritime transport in bulk according to IMO instruments**

not applicable

**Other applicable information**

No dangerous good in sense of these transport regulations.

## SECTION 15: Regulatory information

**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture**

**EU regulatory information**

Restrictions on use (REACH, annex XVII):

Entry 3

**National regulatory information**

Water hazard class (D): 1 - slightly hazardous to water

**15.2. Chemical safety assessment**

Chemical safety assessments for substances in this mixture were not carried out.

## SECTION 16: Other information

**Changes**

This data sheet contains changes from the previous version in section(s): 2,5,7,8,9,10,11.

**Abbreviations and acronyms**

For abbreviations and acronyms, see: ECHA Guidance on information requirements and chemical safety assessment, chapter R.20 (Table of terms and abbreviations).
ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road; ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ASTM - American Society for the Testing of Materials; ATE - Acute Toxicity Estimates; bw - Body weight; CAO - Cargo Aircraft Only; CAS - Chemical Abstracts Service; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DNEL - Derived No-Effect Level; DOT - Department of Transportation; DSL - Domestic Substances List (Canada); EG - European Union; EN - European standards; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; HMIS - Hazardous Materials Identification System; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISO - International Organisation for Standardization; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MSHA - Mine Safety and Health Administration; n;o;s; - Not Otherwise Specified; NFPA - National Fire Protection Association; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NTP - National Toxicology Program; OECD - Organization for Economic Co-operation and Development; PBT - Persistent, Bioaccumulative and Toxic substance; (Q)SAR - (Quantitative) Structure Activity Relationship;

RCRA - Resource Conservation and Recovery Act; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID -Regulation concerning the International Carriage of Dangerous Goods by Rail; RQ - Reportable Quantity; SADT - Self- Accelerating Decomposition Temperature; SARA - Superfund Amendments and Reauthorization Act; SDS - Safety Data Sheet; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

**Classification for mixtures and used evaluation method according to GB CLP Regulation**

| Classification | Classification procedure |
|---|---|
| Acute Tox. 4; H302 | Calculation method |
| Repr. 1B; H360D | Calculation method |
| STOT RE 2; H373 | Calculation method |

**Relevant H and EUH statements (number and full text)**

H302 Harmful if swallowed.
H360D May damage the unborn child.
H361d Suspected of damaging the unborn child.
H373 May cause damage to organs through prolonged or repeated exposure. H411 Toxic to aquatic life with long lasting effects.

**Further Information**

The mixture is classified as hazardous according to regulation (EC) No 1272/2008 [CLP].

The above information describes exclusively the safety requirements of the product and is based on our present-day knowledge. The information is intended to give you advice about the safe handling of the product named in this safety data sheet, for storage, processing, transport and disposal. The information cannot be transferred to other products. In the case of mixing the product with other products or in the case of processing, the information on this safety data sheet is not necessarily valid for the new made-up material.

The receiver of our product is singularly responsible for adhering to existing laws and regulations.

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*(The data for the hazardous ingredients were taken respectively from the last version of the sub-contractor's safety data sheet.)*